Foucault Pendulum

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Background: The Foucault pendulum is a classical demonstration of Earth's rotation. In an inertial (non-rotating) frame, a simple pendulum exhibits simple harmonic motion confined to a fixed plane. However, when analyzed in Earth's rotating reference frame, additional inertial forces arise. In particular, the Coriolis force introduces velocity-dependent coupling between the horizontal components of motion. This coupling causes the oscillation plane of the pendulum to precess over time at a rate proportional to Earth's angular velocity and the sine of the latitude, $\Omega sin\emptyset$. To model it, the second-order differential equation has to be rearranged into a one-order system, so it can be written in Python programming for accuracy and stability. The final modelling provided an effective way to compute the system behavior across different latitudes.

Description: The motion of the pendulum in a rotating frame is described by coupled second-order differential equations derived from Newton's second law, including Coriolis terms. To enable numerical simulation, these equations were reformulated as a system of first-order ordinary differential equations by introducing velocity variables:

$$\dot{x} = v_x, \dot{y} = v_y$$
$$\dot{v_x} = -\Omega_0^2 x + 2\Omega sin\emptyset v_y$$
$$\dot{v_y} = -\Omega_0^2 y - 2\Omega sin\emptyset v_x$$

equation: first-order system

This transformation allows the system to be solved using numerical time-stepping methods. An Euler method was implemented to integrate the equations over time, providing stable and accurate approximations of the pendulum's trajectory. Also, the Runge–Kutta method is a good choice for further accuracy in computation. The simulation was based on the Python library NumPy for numerical computation and Matplotlib for visualization. Animation was introduced using NumPy. Animation inside the library and play as an HTML format in the JupyterNotebooks for real-time observation.

Applications: The Foucault pendulum provides a direct physical demonstration of Earth's rotation and illustrates the role of non-inertial reference frames in classical mechanics. Beyond pendulum motion, the same Coriolis principles govern large-scale atmospheric and oceanic circulation, influencing weather systems, trade winds, and ocean currents.
From a computational perspective, this project demonstrates how a physics system can be described by a differential equation and how we transformed it into a way that a computer can understand. This project also applies Euler's method, which we learned in lecture, to implement the differential equation for the Foucault Pendulum. This method is widely used in all kinds of scientific areas for modelling.

References:

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